Introduction to Linear Programming

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Overview

Today, we cover:

- Talk about final project proposals
- Linear programming

Announcements

- Sign up for final project proposals slot

Introduction to LP

Linear programming (LP) or **linear optimization** is a set of (constrained) optimization algorithms.

- began in 1947 when George Dantzig devised the simplex method
- LP and **quadratic programming** are two special cases of convex optimization with linear constraints

Widely used across industries!

Linear programming

An optimization problem with a linear objective function and linear constraint functions is called a linear program (LP):

$$\begin{aligned} \text{minimize } & c^T x + d \\ \text{subject to } & Ax \preceq b \end{aligned}$$

- **Recall**: you can always convert a minimization problem to a maximization problem by multiplying the objective function by -1
- In statistics, applications include LASSO, quantile regression, support vector machine

Linear programming techniques

Methods for solving LP problems:

- Graphical method
- Simplex algorithm
- Interior point methods

Linear programming techniques

rq {quantreg} R Documentation

Quantile Regression

Description

Returns an object of class `"rq"` `"rqs"` or `"rq.process"` that represents a quantile regression fit.

`method` the algorithmic method used to compute the fit. There are several options:

1. `"br"` The default method is the modified version of the Barrodale and Roberts algorithm for l_1-regression, used by `l1fit` in S, and is described in detail in Koenker and d'Orey(1987, 1994), default = `"br"`. This is quite efficient for problems up to several thousand observations, and may be used to compute the full quantile regression process. It also implements a scheme for computing confidence intervals for the estimated parameters, based on inversion of a rank test described in Koenker(1994).
2. `"fn"` For larger problems it is advantageous to use the Frisch–Newton interior point method `"fn"`. This is described in detail in Portnoy and Koenker(1997).

Carpenter problem

A carpenter can make tables or she can make bookshelves.

- **Table**: costs 10 units of lumber and 5 hours of labor
  - Provides $180 in profit
- **Bookcase**: costs 20 units of lumber, 4 hours of labor
  - $200 in profit
- Only 200 units of lumber available
- 80 hours of labor available

What number of tables and bookcases provide the optimal profit?

Carpenter problem

- **Table**: costs 10 units of lumber and 5 hours of labor, $180 in profit
- **Bookcase**: costs 20 units of lumber, 4 hours of labor, $200 in profit
- 200 units of lumber available; 80 hours of labor available

Let x_1 and x_2 denote number of tables, bookshelves respectively - $10x_1 + 20x_2 \leq 200$ - $5x_1 + 4x_2 \leq 80$ - $x_1, x_2 \geq 0$ - maximize $f(x_1, x_2) = 180x_1 + 200x_2$

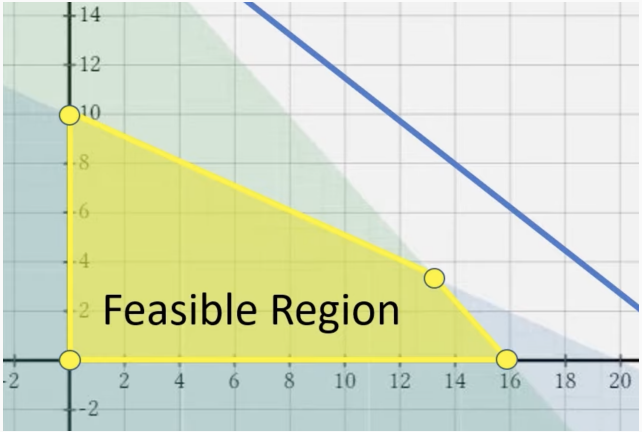
Carpenter problem

Solving LP graphically

The small 2-variable problem can be represented in a 2D graph. The isoprofit line (any point on the line generate the same profit) is perpendicular to the gradient of objective function. The problem can be solved by sliding the isoprofit line.

Feasible region

Shaded region that satisfies constraints is called feasible region



Properties of the optimal solution

Optimal solution might not exist

- the objective function is unbounded in the solution space (can be improved indefinitely without violating any constraints)
- the feasible region is empty (no solution satisfies all constraints)

But if the optimal solutions exists:

- Interior point: No.
- Corner point: Yes. The corner points are often referred to as "extreme points" or vertices.
- Edge point: Yes, only when the edge is parallel to the isoprofit line. In this case, every point on that edge has the same objective function value, including the corner point of that edge.
- **Key result**: If an LP has an optimal solution, it must have a extreme point optimal solution. This greatly reduces our search of optimal solutions: only need to check a finite number of extreme points.

Big ideas of LP

- Number of vertices in feasible regions corresponds to number of constraints
- inequalities make a bounded, concave feasible region
- Extreme values (min and max) occur at the vertices of this region
- Solution: compute vertices, choose which one gives largest value of $f(x_1, x_2)$

You can solve this algorithmically (rather than graphically) using the simplex method.

Simplex method

Linear programming involves optimizing (maximizing or minimizing) a linear objective function, subject to a set of linear constraints. The Simplex method helps to find the optimal solution to such problems, provided one exists.

For simplex method:

- the LP problem needs to be in **standard** form
  - objective function set up as a maximization problem
  - constraints set up as linear inequalities of the form $Ax \leq b$
  - need all variables x to be nonnegative

LP Standard form

An LP can be expressed using only equality and nonnegativity constraints. This is called **standard form**:

$$\begin{aligned} \text{minimize } & c^T x \\ \text{subject to } & Ax = b \\ & x \geq 0 \end{aligned}$$

An LP can also be expressed using only inequality constraints. This is called **inequality form**:

$$\begin{aligned} \text{minimize } & c^T x \\ \text{subject to } & Ax \leq b \end{aligned}$$

Slack variables

Are added to less than or equal to inequality constraints to transform them into equality constraints

- represent the "unused" capacity within a constraint
- non-negative variables
- Say you have inequality $2x + 3y \leq 10$.
  - Convert to to equality constraint $2x + 3y + s = 10$
  - If $s \neq 0$, then there the combination of x and y could be increased further without violating the constraint

Simplex: convert to standard form using slack variables

Go back to Carpenter example.

- maximize $z = f(x_1, x_2) = 180x_1 + 200x_2$
- $x_1, x_2 \geq 0$

Constraints:

- $10x_1 + 20x_2 \leq 200 \rightarrow 10x_1 + 20x_2 + s_1 = 200$
- $5x_1 + 4x_2 \leq 80 \rightarrow 5x_1 + 4x_2 + s_2 = 80$

Slack variables pick up any "slack" in the system based on current values of x_1 and x_2

- $s_1, s_2 \geq 0$
- $s_1, s_2 = 0$ when z is maximized

Steps of Simplex algorithm

1. Convert problem to standard form
2. Set up "tableau": basically amounts to writing out problem as a matrix
3. Start at a vertex of the feasible region, then pivot

- Essentially, you are iteratively moving from one vertex (or "corner point") of the feasible region to another, with the goal of improving the objective function at each step.

Initial tableau

First $z = 180x_1 + 200x_2 \rightarrow z - 180x_1 - 200x_2 = 0$

- $10x_1 + 20x_2 \leq 200 \rightarrow 10x_1 + 20x_2 + s_1 = 200$
- $5x_1 + 4x_2 \leq 80 \rightarrow 5x_1 + 4x_2 + s_2 = 80$

Start at $x_1, x_2 = 0$

Pivot step

Pivot step

Pivot step

Solution

Summary of the Simplex method

Basic variables define current corner point/vertex!

- Those are the variables that are literally a unit basis of the matrix
- Walks along the edge of the feasible space to improve the objective function
  - "Basic" variables are the current solution
- Only considers vertices
- Moves as far as it can in optimal direction until movement is blocked by another constraint

Standard form as a matrix

$$\begin{aligned} \max z &= c^T x \\ s.t. Ax &= b \\ x &\geq 0 \end{aligned}$$

$$x = [x_1, x_2, s_1, s_2]^T$$

Linear programming

An optimization problem with a linear objective function and linear constraint functions is called a linear program (LP):

$$\begin{aligned} \text{minimize} \quad & c^T x + d \\ \text{subject to} \quad & Ax \preceq b \end{aligned}$$

Linear programming techniques

- Graphical method
  - Only works for 2 variables
- Simplex algorithm
  - Moves from one basic feasible solution to the next
  - Standard version works for $\leq$ constraints
- Two phase, big M methods
  - Variations on Simplex method
  - Handle $=$ and $\geq$ constraints
- Interior point methods
  - Better for large-scale LP problems
  - Instead of moving along boundary, moves through interior of feasible region

Interior point methods

Effectively, the idea is to set up the problem such that we can solve it using something like Newton's method.

- Need to set up some background and theory first
- Allows us to handle the constraints

Carpenter problem

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- Only 200 units of lumber available
- 80 hours of labor available

What number of tables and bookcases provide the optimal profit?

LP problem for resource allocation

Take a typical maximizing LP problem (3 variables and 2 constraints).

$$
\begin{aligned}
\max \quad & c_1x_1 + c_2x_2 + c_3x_3 \\
\text{s.t.} \quad & a_{11}x_1 + a_{12}x_2 + a_{13}x_3 \leq b_1 \\
& a_{21}x_1 + a_{22}x_2 + a_{23}x_3 \leq b_2 \\
& x_1, x_2, x_3 \geq 0
\end{aligned}
$$

Economic interpretation of the problem:

- We produce three products using two materials - x_j: unit of

Resource valuation problem

Now, suppose a buyer is considering purchasing our entire inventory of materials but is unsure how to price them. The buyer knows we will only agree to the sale if it yields a higher return than using the materials for production.

Buyer's business strategy: producing one fewer unit of product j will save us:

- a_{1j} units of material 1 and a_{2j} units of material 2

The buyer seeks to determine the unit prices of materials to minimize costs while ensuring we still agree to the sale (i.e., we do not earn less). Let y_1 and y_2 be the unknown unit prices of the

Resource valuation problem

$$
\begin{aligned}
\max \quad & c_1x_1 + c_2x_2 + c_3x_3 \\
\text{s.t.} \quad & a_{11}x_1 + a_{12}x_2 + a_{13}x_3 \leq b_1 \\
& a_{21}x_1 + a_{22}x_2 + a_{23}x_3 \leq b_2 \\
& x_1, x_2, x_3 \geq 0
\end{aligned}
$$

$$
\begin{aligned}
\min \quad & b_1y_1 + b_2y_2 \\
\text{s.t.} \quad & a_{11}y_1 + a_{21}y_2 \geq c_1 \\
& a_{12}y_1 + a_{22}y_2 \geq c_2 \\
& a_{13}y_1 + a_{23}y_2 \geq c_3 \\
& y_1, y_2 \geq 0
\end{aligned}
$$

Duality

The buyer's LP problem is called the "**dual**" problem of the original problem, which is called the "**primal** problem.

In matrix notation, if the **primal** LP problem is:

$$\begin{aligned} \max \quad & cx \\ \text{s.t.} \quad & Ax \leq b, x \geq 0 \end{aligned}$$

The corresponding **dual** problem is:

Duality

We can also express the dual problem in canonical form (a maximization problem with $\leq$ constraints):

$$\begin{aligned} \max \quad & -b^T y \\ \text{s.t.} \quad & -A^T y \leq -c^T, y \geq 0 \end{aligned}$$

The **dual is the negative transpose of the primal**. - This means that the dual of the dual problem is the primal problem

Why do we care about duality?

Duality

We can also express the dual problem in canonical form (a maximization problem with $\leq$ constraints):

$$\begin{aligned} \max \quad & -b^T y \\ \text{s.t.} \quad & -A^T y \leq -c^T, y \geq 0 \end{aligned}$$

Why do we care about duality? - Sometimes the dual problem is easier to solve than the primal one - If a constraint is slightly relaxed or tightened, the dual variables indicate how much the objective function will change - Solution to primal is **always** $\leq$

Duality in non-canonical form

What if the primal problem doesn't fit into the canonical form (e.g., has $\geq$ or $=$ constraints, unrestricted variables)?

The **variable types in the dual problem** are determined by the **constraint types in the primal problem** as follows:

- Equality constraints in the primal $\rightarrow$ Unrestricted dual variables
- $\leq$ in the primal $\rightarrow$ **nonnegative** dual variables
- $\geq$ in the primal $\rightarrow$ **nonpositive** dual variables

i.e.

Duality in non-canonical form

Conversely, **constraint types of the dual** problem are determined by **variable types of the primal**:

Primal (max) variable	**Dual (min) constraints**
≥ 0	$\geq$
≤ 0	$\leq$
unrestricted	$=$

Non-canonical duality example

If the primal problem is:

$$
\begin{aligned}
max \quad & 20x_1 + 10x_2 + 50x_3 \\
s.t. \quad & 3x_1 + x_2 + 9x_3 \leq 10 \\
& 7x_1 + 2x_2 + 3x_3 = 8 \\
& 6x_1 + x_2 + 10x_3 \geq 1 \\
& x_1 \geq 0, \quad x_2 \textit{ unrestricted}, \quad x_3 \leq 0
\end{aligned}
$$

The dual problem is:

Non-canonical duality example

If the primal problem is:

$$
\begin{aligned}
max \quad & 20x_1 + 10x_2 + 50x_3 \\
s.t. \quad & 3x_1 + x_2 + 9x_3 \leq 10 \\
& 7x_1 + 2x_2 + 3x_3 = 8 \\
& 6x_1 + x_2 + 10x_3 \geq 1 \\
& x_1 \geq 0, \quad x_2 \textit{ unrestricted}, \quad x_3 \leq 0
\end{aligned}
$$

The dual problem is:

Weak duality

Weak duality theorem:

- For any feasible solution x to the **primal** (maximization) problem and any feasible solution y to the **dual** (minimization) problem:

$$\text{objective of dual} \;\geq\; \text{objective of primal.}$$

- Implications:
  - The dual provides a bound on the primal's optimal value.
  - Useful for proving infeasibility or bounds in optimization problems.
- **Weak Duality Guarantee**: the optimal value of the primal

Weak duality proof

Weak duality: If $(x_1, \dots, x_n)$ is a feasible solution for the primal, and $(y_1, \dots, y_m)$ is a feasible solution for the dual, then $\sum_j c_j x_j \leq \sum_i b_i y_i$..

Weak duality proof

Weak duality: If $(x_1, \dots, x_n)$ is a feasible solution for the primal, and $(y_1, \dots, y_m)$ is a feasible solution for the dual, then $\sum_j c_j x_j \leq \sum_i b_i y_i$..

$$\sum_j c_j x_j \leq \sum_j \left(\sum_i y_i a_{ij} \right) x_j \tag{1}$$

$$= \sum_{i,j} y_i a_{ij} x_j \tag{2}$$

$$= \sum_i \left(\sum_j a_{ij} x_j \right) y_i \tag{3}$$

Strong duality

- We know that at feasible solutions for both dual and primal, the solution to the dual is always greater or equal
- If there is a difference between the largest primal value and the smallest dual value it is called the "**Duality gap**".
  - When are these values equal?

Strong Duality Theorem: If the primal has an optimal solution, then the dual also has an optimal solution and there is no duality gap, i.e.:

$$\text{objective of dual} \;=\; \text{objective of primal.}$$

Complementary slackness theorem

The **complementary slackness theorem** provides a necessary condition for optimality in linear programming. It states that:

If x^* and y^* are feasible solutions of primal and dual problems, then x^* and y^* are both optimal if and only if

1. $y^{*T}(b - Ax^*) = 0$
2. $(y^{*T}A - c)x^* = 0$

This theorem is the foundation of the **interior point method** class of LP solvers.

Complementary slackness theorem

Consider a simple LP problem:

$$\begin{aligned} \textit{max} \quad & z = x_1 + x_2, \\ \textit{s.t.} \quad & x_1 + 2x_2 \leq 100, \\ & 2x_1 + x_2 \leq 100, \\ & x_1, x_2 \geq 0. \end{aligned}$$

Its dual problem is:

Complementary slackness theorem

Consider a simple LP problem:

$$
\begin{aligned}
\textit{max} \quad & z = x_1 + x_2, \\
\textit{s.t.} \quad & x_1 + 2x_2 \leq 100, \\
& 2x_1 + x_2 \leq 100, \\
& x_1, x_2 \geq 0.
\end{aligned}
$$

Its dual problem is:

Complementary slackness theorem

The complementary slackness theorem states...

Complementary slackness theorem

The complementary slackness theorem states:

$$
\begin{aligned}
y_1(100 - x_1 - 2x_2) &= 0, \\
y_2(100 - 2x_1 - x_2) &= 0, \\
x_1(y_1 + 2y_2 - 1) &= 0, \\
x_2(2y_1 + y_2 - 1) &= 0.
\end{aligned}
$$

At the optimal point, we have $\mathbf{x} = (\mathbf{100/3}, \mathbf{100/3})$, $\mathbf{y} = (\mathbf{1/3}, \mathbf{1/3})$. The complementary slackness holds.

- All the constraints are bounded in both primal and dual

Plot of constraints

Complementary slackness theorem

Modify the simple LP problem a bit:

$$
\begin{aligned}
\textit{max} \quad & z = 3x_1 + x_2, \\
\textit{s.t.} \quad & x_1 + 2x_2 \leq 100, \\
& 2x_1 + x_2 \leq 100, \\
& x_1, x_2 \geq 0.
\end{aligned}
$$

and its dual problem is:

Complementary slackness theorem

Now the complementary slackness theorem states:

$$
\begin{aligned}
y_1(100 - x_1 - 2x_2) &= 0, \\
y_2(100 - 2x_1 - x_2) &= 0, \\
x_1(y_1 + 2y_2 - 3) &= 0, \\
x_2(2y_1 + y_2 - 1) &= 0.
\end{aligned}
$$

For this LP problem, the optimal solutions are $\mathbf{x} = (\mathbf{50}, \mathbf{0})$, $\mathbf{y} = (\mathbf{0}, \mathbf{1.5})$. Complementary slackness still holds. We observe that:

Complementary slackness theorem

For this LP problem, the optimal solutions are $\mathbf{x} = (\mathbf{50}, \mathbf{0})$, $\mathbf{y} = (\mathbf{0}, \mathbf{1.5})$. Complementary slackness still holds. We observe that:

- In the primal problem:
  - The first constraint is unbounded, so its corresponding variable in the dual problem (y_1) must be 0.
  - Second constraint is bounded, so y_2 can be non-zero.
- In the dual problem:
  - The first constraint is bounded, so its corresponding variable in the primal problem (x_1) can be non-zero.
  -

Plot of constraints

Economics interpretation

Dual variables can be called "**shadow prices**" of the primal constraint, how objective function would increase if constraint was relaxed.

- If a primal constraint is bounded, relaxing that constraint would result in a gain (improve the objective function), shadow price is non-zero.
- If a primal constraint is unbounded, relaxing that constraint would not improve the objective function, shadow price is zero.

Economics interpretation

Bakery example: Imagine you're running a bakery, and can make a limited number of cakes because you have a limited amount of flour. Shadow price tells you how much more profit you'd gain if you had a little more flour.

- If you're tight on flour, getting more flour would let you bake more cakes and make more money. - In this case, the shadow price is nonzero because the extra resource improves your outcome.
- If you have plenty of flour, adding more doesn't help—you're already making as many cakes as you can sell. In this case, the

Primal/dual optimality conditions

Given the primal and dual problem with slack/surplus variables added:

Primal:

$$\begin{aligned} \textit{max} \quad & cx \\ \textit{s.t.} \quad & Ax + w = b, \quad x, w \geq 0. \end{aligned}$$

Dual:

$$\begin{aligned} \textit{min} \quad & b^T y \\ \textit{s.t.} \quad & A^T y - z = c^T, \quad y, z \geq 0. \end{aligned}$$

Primal/dual optimality conditions

We now have the optimality conditions for the primal/dual problems as:

$$
\begin{aligned}
Ax + w - b &= 0, \\
A^T y - z - c^T &= 0, \\
XZe &= 0, \\
WYe &= 0, \\
x, y, w, z &\geq 0.
\end{aligned}
$$

- The first two conditions are simply the constraints for

Primal/dual optimality conditions

We now have the optimality conditions for the primal/dual problems as:

$$
\begin{aligned}
Ax + w - b &= 0, \\
A^T y - z - c^T &= 0, \\
XZe &= 0, \\
WYe &= 0, \\
x, y, w, z &\geq 0.
\end{aligned}
$$

Ignoring the non-negativity constraints, this is a set of $2n + 2m$

Primal-dual interior point method

The primal-dual interior point method finds the primal-dual optimal solution (x^*, y^*, w^*, z^*) by applying Newton's method to the primal-dual optimality conditions.

- The direction and length of the steps are modified in each step so that the non-negativity condition is strictly satisfied in each iteration.

To be specific, define the following function $F : R^{2n+2m} \to R^{2n+2m}$:

$$\begin{bmatrix} \mathbf{A}x + \mathbf{w} - \mathbf{b} \end{bmatrix}$$

Primal-dual interior point method

Applying Newton's method, if at iteration k the variables are (x^k, y^k, w^k, z^k), we obtain a search direction $(\delta x, \delta y, \delta w, \delta z)$ by solving the linear equations:

$$F'(x^k, y^k, w^k, z^k) \begin{bmatrix} \delta x \\ \delta y \\ \delta w \\ \delta z \end{bmatrix} = -F(x^k, y^k, w^k, z^k).$$